

DBMS – ASSIGNMENT 1

BSDS, Second Year, 2025–2026

Deadline: February 15, 2026

Total: 5 marks

SUBMISSION INSTRUCTIONS

1. Submit all the solutions in a single file.
2. Naming convention for your submission file (assuming BSD-BG/CC/DD-24xx is your roll number): BSD-BG/CC/DD-24xx-assign1 (.rtf, .docx, .doc, .pdf, .tex, etc.).
3. To submit a solution file (say BSD-BG/CC/DD-24xx-assign1.pdf), ensure that it is not password protected and mail to <assignisik@gmail.com> with the subject line as follows: DBMS BSDS 2024-28 BSD-BG/CC/DD-24xx Assignment 1.

NOTE: All the solutions must be self-sufficient and to the point.

Q1. For representing regular expressions, **re** library can be used in python. Write a program to verify whether a phone number inserted to a database is valid or not based on the following requirements.

- It comprises of digits 0 to 9, and dash only.
- It has a length of 12.
- It has a STD part of length 3 and a local part of length 8, separated by a dash.

Q2. Let there be three relations $R_1(V, W, X)$, $R_2(W, X, Y)$ and $R_3(X, Y, Z)$ each having $n > 0$ tuples. Consider that V , W , X , Y and Z take integer values only. Suppose the operations $\pi_X(R_1)$, $\pi_X(R_2)$ and $\pi_X(R_3)$ return t_1 , t_2 and t_3 tuples, respectively. Without making any further assumptions, find out the minimum and maximum possible number of tuples that may appear in the resulting relations provided by the following operations.

- (i) $R_1 \bowtie R_2 \bowtie R_3$
- (ii) $\pi_X(R_1 \bowtie R_2 \bowtie R_3)$
- (iii) $\pi_X(R_1) \cup \pi_X(R_2) \cup \pi_X(R_3)$
- (iv) $(R_1 \ltimes R_2) \cup (R_1 \rhd R_2)$